

Marathi → English Learning App

Complete Project Specification – v2.0

Base plan: Narendra · Enhanced with gap-analysis additions (marked )

Changelog – What's New in v2.0

Original plan already covered vision, structure, content design, and phased development well. This version adds 10 items that were missing from a real-user / real-device standpoint:

1.  Spaced Repetition System (SRS) for actual vocabulary retention
2.  Audio strategy reality-check – device TTS coverage risk + bundled fallback pack
3.  Offline vs. online reality-check for Speech Recognition
4.  Content translation QA / native-speaker review pipeline
5.  Daily reminder notifications (habit-building)

6.  Content update mechanism without full app updates
 7.  Pure audio "Listen & Repeat" mode for weak/non-readers
 8.  Play Store compliance basics (privacy policy, permissions, data safety)
 9.  Low-end device & performance testing plan
 10.  Growth/virality feature (WhatsApp streak sharing)
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1. Project Vision

A completely free Android English-learning app built specifically for Marathi-speaking beginners who have zero or very weak English — weak vocabulary, weak grammar, and fear of speaking.

Goal progression:

"English येत नाही" → "English समजते" → "English वाचता येते" → "Basic English बोलता येते"

The app must feel simple, friendly, non-judgmental, and usable by someone who has never studied English properly.

2. Core Learning Philosophy

No heavy grammar-first teaching. Learning flows naturally:

Marathi Meaning → English Word →
Pronunciation → Example Sentence
→ Audio → Practice → Quiz →
Speaking

3. Languages

- **Primary learning language:** Marathi
- **Support languages:** Hindi, English

Every vocabulary item shows English + Hindi + Marathi + Pronunciation, e.g.:

English	Hindi	Marathi	Pronunciation
Water	पानी	पाणी	वॉटर

4. App Structure (Navigation)

1. Home
2. Learn
3. Vocabulary
4. Practice

- 5. Speaking
 - 6. Daily Challenge
 - 7. My Words / Favorites
 - 8. Progress
 - 9. Settings
-

5. Home Screen

Simple, greeting-based, action-first.

- Greeting: "Good Morning / Good Evening" + "आज 5 मिनिटे English शिकूया!"
 - Today's snapshot: 5 new words, 3 sentences, speaking practice, daily quiz, streak, overall progress
 - Single primary CTA: **"Start Today's Learning"**
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6. Learn Section

6.1 Level 1 — Beginner

Alphabet, basic words, numbers, colors, family, body parts, food, home, common objects.

Example: घर → House → मकान → हाऊस

6.2 Level 2 — Daily Vocabulary Categories

Home, Family, Food, Office, School, College, Travel,
Railway, Bus, Shopping, Hospital, Bank, Mobile, Internet,
Time, Weather, Emotions, Common Actions.

6.3 Level 3 – Daily Sentences

Practical, everyday sentences, e.g.:

- मला पाणी पाहिजे. → I want water.
 - मला समजले नाही. → I didn't understand.
 - तुम्ही मला मदत करू शकता का? → Can you help me?
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7. Vocabulary Library

Searchable, revisable library. Each word entry shows:
English, Hindi, Marathi, pronunciation, word type,
V1/V2/V3/V-ing (for verbs), example sentences, audio,
and a favorite toggle.

Example card – **Speak**: बोलना (Hindi) / बोलणे (Marathi) /
स्पीक (pronunciation) / Verb – V1 Speak, V2 Spoke, V3
Spoken, V-ing Speaking, with example sentences in all
three languages and a  Listen +  Add to My Words
action.

8. Verb Library (V1 / V2 / V3)

Dedicated section for common verbs, not just a table – each verb shown with real sentence usage.

English	Hindi	Marathi	V1	V2	V3
Go	जाना	जाणे	Go	Went	Gone
Eat	खाना	खाणे	Eat	Ate	Eaten
Speak	बोलना	बोलणे	Speak	Spoke	Spoken

9. Search

Search works across English, Hindi, and Marathi input and returns the matching word card regardless of which language was typed.

10. Filters & Categories

Filters: A-Z, Most Important, Recently Viewed, Favorites, Verbs, Nouns, Adjectives, Daily Words, Common Words.

Categories: Home, Family, Food, Office, Travel, Shopping, Hospital, Education, Technology, Money, Time, Weather, Emotions, Common Actions, Daily Conversation.

11. My Words / Favorites

- Saved Words, Recently Viewed, Difficult Words, Learned Words
 - Remove-from-favorites and "practice saved words" actions
 - **Difficult Words** should auto-populate from repeated wrong answers (feeds directly into the SRS system in Section 16)
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12. Pronunciation & Audio Strategy



Original plan: rely on device/free Text-to-Speech, avoid paid APIs.

Reality-check: Marathi TTS voice packs are inconsistent across Android OEMs — many budget/older phones only ship Hindi or English voices, not Marathi. Since the target audience skews toward budget devices, this is a real risk to the core feature.

Mitigation:

- Use device TTS as the default (works fine for English words)
- Bundle a small **human-recorded audio pack** (MP3, ~5–10 MB) for the highest-priority ~500

words/phrases as a guaranteed fallback,
independent of device TTS support

- At first launch, silently test whether Marathi TTS is available and fall back automatically if not
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13. Practice Section

Modes: Vocabulary Quiz (MCQ), Marathi → English translation, English → Marathi translation, Word Matching.

14. Speaking Practice

Original plan: use device-supported speech recognition, keep scoring simple in V1.

Reality-check: Android's on-device `SpeechRecognizer` for Indian-accented English generally needs an internet connection and the Google app — this conflicts directly with the "offline-first" goal.

Mitigation:

- Label Speaking Practice clearly as an **online feature** in V1, OR
- Ship a simpler **offline fallback**: user records their voice, plays it back next to the model audio, and

self-rates (no scoring engine needed) — keeps the offline promise intact for the core app

15. Daily Challenge

Each day: 5 new words, 3 new sentences, 1 quiz, 1 speaking practice — auto-curated from the current level.

16. Spaced Repetition System (SRS)

 — New

Why it's needed: Quizzes and a static Favorites list don't create long-term retention. Beginners will forget words within days without a review schedule — this is the single biggest gap in the original plan.

Proposed design (Leitner-style, simple enough for V1):

- 4 review boxes: New → Learning (1 day) → Reviewing (3 days) → Mastered (7+ days)
- Correct answer in Practice/Quiz → word moves to the next box
- Wrong answer → word drops back to "Learning" and also tags into Difficult Words

- Home screen's "Today's Learning" pulls due reviews from this queue automatically, mixed with new words

This turns the app from a vocabulary list into an actual retention system.

17. Grammar Through Sentence Patterns

No isolated grammar rules — patterns taught through repetition:

```
I am... → I am happy. / I am tired. / I am busy.  
I have... → I have a phone. / I have work.  
I want... → I want water. / I want help.  
I need... → I need help. / I need time.
```

18. Real-Life English Modules

Situational phrase sets: At Home, At Office, Railway Station, Shopping, Hospital, Travel — each with 4-6 ready-to-use practical sentences.

19. Progress Tracking

Simple counters, no complex dashboards: Vocabulary (X/500), Sentences (X/300), Lessons (X/50), Speaking (X/30), Quiz average %.

20. Streak & Daily Reminders

Original plan: track a learning streak, keep it low-pressure, not stressful.

Addition: A streak alone doesn't bring users back — it only rewards them *after* they return. Add a **local daily reminder notification** (no server needed, fully offline-compatible) at a user-chosen time:

"आज तुझा 5 मिनिटांचा अभ्यास बाकी आहे 

This is the single highest-leverage retention feature for a habit-building app like this.

21. Content Update Mechanism — New

With an offline-first, local-DB architecture, shipping new vocabulary/lessons normally requires a full app update. Instead:

- Version the vocabulary/lesson JSON content separately from the app binary

- On Wi-Fi + user opt-in, allow downloading incremental "content packs" (new categories, more words) without a Play Store update
 - Keeps the app light initially and lets content grow toward the 500+/300+ targets over time
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22. Content Quality & Translation

Verification — New

The original plan (Section 29 equivalent) already flags "no unnatural literal translations" as a rule — but there's no process to enforce it.

Addition: Before any content batch ships, run it through a native Marathi-speaker review pass (even informally) to catch literal/awkward translations. This matters more for trust than any single feature — one bad translation undermines the whole "app that understands beginners" positioning.

23. Offline-First Architecture

Core learning content (vocabulary, sentences, lessons, quizzes, favorites, progress) stored locally via SQLite/local JSON. No mandatory backend for the basic experience.

24. No Login in Version 1

First launch → pick Marathi/Hindi → start learning immediately. All progress stored locally, no forced account creation.

25. Free-First Approach

No paid API dependency, no paid AI dependency, no mandatory cloud backend, no mandatory login, no subscription for the core experience. AI features are optional and deferred to V2.

26. Accessibility (expanded)

Original: large fonts, audio buttons, simple icons, high readability, minimal text density.

Addition: Some target users are not fully comfortable reading Marathi either. Add a **"Listen & Repeat" audio-only mode** for Level 1 — no reading required, just hear-and-repeat — so literacy isn't a hard gate to entry. Also ensure TalkBack labels are present on all icon-only buttons, and support system font-scaling.

27. Onboarding Flow

1. "English शिकायला सुरुवात करा" + subtitle "मराठीतून सोप्या पद्धतीने English शिका."
 2. Ask current level (4 options, from "अजिबात येत नाही" to "बोलण्याचा सराव हवा आहे")
 3. Recommend a starting level based on the answer
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28. UI/UX Design Principles

Clean, modern, beginner-friendly, large readable text, simple navigation, minimal animations, Marathi-first UI copy (with a language toggle for the UI chrome itself, not just content, so Hindi-comfortable users aren't stuck with Marathi menus).

29. Data Structure (Vocabulary Schema)

```
{
  "word": "Go",
  "hindi": "जाना",
  "marathi": "जाणे",
  "pronunciation": "गो",
  "type": "verb",
  "v1": "Go",
  "v2": "Went",
  "v3": "Gone",
  "ving": "Going",
```

```
"srs_box": "new",
"examples": [
  {
    "english": "I go to work.",
    "hindi": "मैं काम पर जाता हूँ।",
    "marathi": "मी कामावर जातो."
  }
]
```

(`srs_box` field added to support Section 16's spaced repetition system.)

30. Initial Content Targets (V1)

- 500+ vocabulary words
 - 200+ common verbs
 - 300+ daily sentences
 - 50+ lessons
 - 500+ quiz questions
 - 20+ real-life situations
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31. Growth & Virality Features NEW — New

A simple, low-effort growth lever well suited to the Indian market: a **"Share to WhatsApp Status"** button on

streak/milestone screens (e.g., "5-Day Streak 🔥 " or "50 Words Learned"). No backend needed — just a shareable image/text generated on-device.

32. Compliance & Play Store Requirements — New

Even with no login, Play Console will require:

- A **Privacy Policy URL** (hosted anywhere, even a simple static page)
 - Justification for the **microphone permission** (Speaking Practice)
 - A completed **Data Safety** declaration form
 - Minimum SDK / target SDK version decisions upfront (affects device reach)
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33. Device & Performance Testing — New

Target users likely skew toward budget/older Android devices. Before launch:

- Test on at least one low-RAM device (2GB or less)
- Track APK size budget (especially once audio packs are bundled — consider Play Feature Delivery)

for optional assets)

- Confirm Marathi/Hindi Devanagari text renders correctly across common OEM skins (Samsung, Xiaomi, Vivo, etc.)

34. Future AI Features – Version 2 (not built initially)

- **AI English Tutor** – simple conversational practice
- **AI Conversation Practice** – situational (interview, shopping, phone calls) with mistake correction explained in Marathi, e.g.: "You is going" → "You are going." – "आपण 'You' सोबत 'are' वापरतो."

35. Development Roadmap (Phases)

Phase	Scope
1 – Foundation	Project setup, navigation, theme, local DB, onboarding, home screen
2 – Learning Content	Lessons, vocabulary, search, categories, word details, V1/V2/V3

Phase	Scope
3 – Practice	Quiz, matching, MR ↔ EN translation, favorites, SRS engine NEW
4 – Audio	TTS integration + bundled fallback audio pack NEW
5 – Progress & Retention	Streak, completed lessons, daily reminder notifications NEW
6 – Speaking	Speech recognition (online) + offline record/compare fallback NEW
7 – Content QA & Compliance NEW	Native-speaker translation review, privacy policy, permissions, data safety form
8 – Testing & Release	Low-end device testing, offline functionality, text rendering, APK build

36. Definition of Done (Before Calling V1 Complete)

- No broken navigation, no placeholder content, no dummy buttons, no empty screens

- No paid API/backend requirement, no mandatory login
 - Works offline for all core features (Speaking clearly labeled if it needs internet)
 - Marathi/Hindi/English text renders correctly across devices
 - Search, Favorites, and Progress persist correctly after closing the app
 - SRS queue correctly surfaces due reviews on Home
 - Daily reminder notification fires reliably
 - Privacy Policy + Data Safety form completed
 - APK installs and runs correctly on a low-end test device
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37. Final Product Positioning

A completely free, Marathi-first English learning app for absolute beginners, built around the idea that "English शिकण्यासाठी आधी English येणे गरजेचे नाही."

Learning approach: **Marathi → English → Practice → Speaking**, with retention (SRS + reminders) and trust (verified translations, encouraging tone) as the two pillars that turn a good content plan into an app people actually keep using.